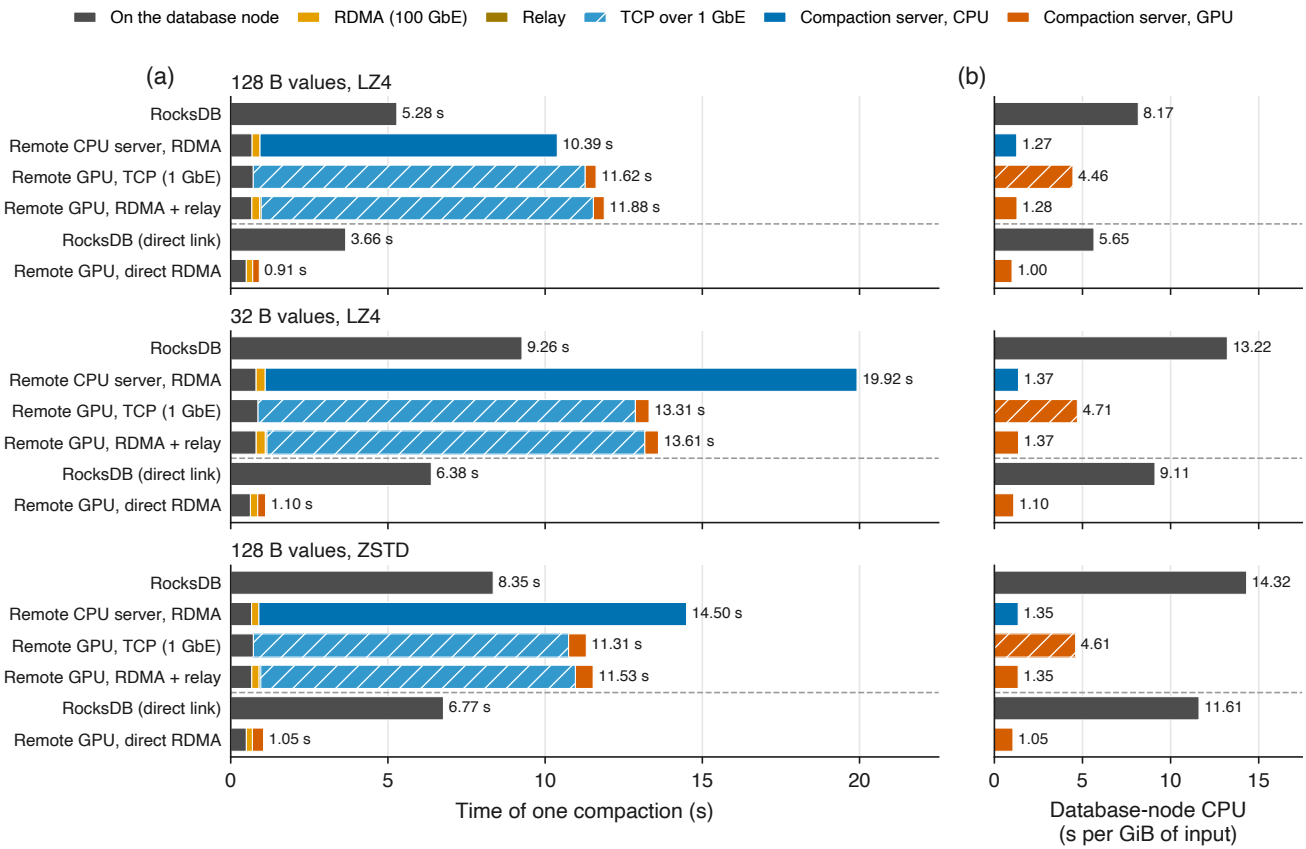
On the database node
RDMA (100 GbE)
Relay
TCP over 1 GbE
Compaction server, CPU
Compaction server, GPU
(a)
128 B values, LZ4
RocksDB
5.28 s
Remote CPU server, RDMA
10.39 s
Remote GPU, TCP (1 GbE)
11.62 s
Remote GPU, RDMA + relay
11.88 s
RocksDB (direct link)
3.66 s
Remote GPU, direct RDMA
0.91 s
32 B values, LZ4
RocksDB
9.26 s
Remote CPU server, RDMA
19.92 s
Remote GPU, TCP (1 GbE)
13.31 s
Remote GPU, RDMA + relay
13.61 s
RocksDB (direct link)
6.38 s
Remote GPU, direct RDMA
1.10 s
128 B values, ZSTD
RocksDB
8.35 s
Remote CPU server, RDMA
14.50 s
Remote GPU, TCP (1 GbE)
11.31 s
Remote GPU, RDMA + relay
11.53 s
RocksDB (direct link)
6.77 s
Remote GPU, direct RDMA
1.05 s
0
5
10
15
20
Time of one compaction (s)
(b)
8.17
1.27
4.46
1.28
5.65
1.00
13.22
1.37
4.71
1.37
9.11
1.10
14.32
1.35
4.61
1.35
11.61
1.05
0
5
10
15
Database-node CPU
(s per GiB of input)